PROPERTY R_∞ FOR THE ARTIN GROUPS OF TYPES $E_6, E_7, E_8, F_4, H_3, H_4$

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ABSTRACT. A group has property R_∞ if every automorphism has infinitely many twisted conjugacy classes. We prove that the Artin groups of spherical types E_6, E_7, E_8, F_4, H_3 and H_4 have property R_∞. Combined with results of Fel'shtyn–Gonçalves, Calvez–Soroko and Soroko–Vaskou, this shows that every irreducible Artin group of spherical type and rank at least two has property R_∞. The proof does not use a description of the automorphism group of the Artin group, which is unknown for these types. It rests on three facts:

- the pure Artin group P is characteristic (Cohen–Paris);
- $H_1(P;\mathbb{Z})$ is the permutation module $\mathbb{Z}[\mathcal{T}]$ on the reflections;
- the automorphism of $W/Z(W)$ induced by an automorphism of A is realised by a permutation of $\mathcal{T}$.

Schur's lemma, applied to a rational, absolutely irreducible constituent of multiplicity one of $\mathbb{Q}[\mathcal{T}]$, then gives eigenvalue 1 on $H_1(P;\mathbb{Q})$ after twisting by a suitable inner automorphism. The representation-theoretic input is certified by exact computer calculations, checked with independent programs. For E_8 the automorphisms of $W/Z(W)$ are determined by hand. The same criterion gives an automorphism-free proof for all types A_n, and, by finite computation, for B_n $(3 \le n \le 7)$ and D_n $(4 \le n \le 7)$. We explain why it fails for some dihedral types.

AI-assistance statement (placeholder; to be completed by the human authors). The mathematical argument, the computer programs and this manuscript were produced with substantial assistance from an AI system (Anthropic's Claude). They were checked by AI-based review only, including one AI "cold referee", and have *not yet* been checked by a human expert. The authors must replace this paragraph with an accurate statement of contributions and verification before any submission or posting.

1. INTRODUCTION

1.1. **Twisted conjugacy and property R_∞.** Let G be a group and $\varphi \in \mathrm{Aut}(G)$. Two elements $x, y \in G$ are *φ-twisted conjugate* if $y = zx\varphi(z)^{-1}$ for some $z \in G$. The number $R(\varphi) \in \{1, 2, \dots\} \cup \{\infty\}$ of equivalence classes is the *Reidemeister number* of φ. The group G has *property* R_∞ if $R(\varphi) = \infty$ for every $\varphi \in \mathrm{Aut}(G)$. See [19, 17] for surveys.

For Artin groups the question goes back to Fel'shtyn and Gonçalves, who proved R_∞ for the braid groups [16]. Further results:

- Juhász treated certain large-type Artin groups [26].
- Dekimpe, Gonçalves and Ocampo treated the pure braid groups [12].
- Calvez and Soroko gave a uniform proof for the spherical types $A_n, B_n, D_4, I_2(m)$, their pure subgroups, and the affine types $\tilde{A}_n, \tilde{C}_n$ [6].
- Soroko and Vaskou treated spherical type D_n, $n \ge 6$, and large hyperbolic-type groups [32].
- Soroko treated many non-spherical Artin and Coxeter groups whose outer automorphisms are length-preserving [31].

The question "Which Artin groups have property R_∞?" is Problem 8.6 of the AIM problem list on Artin groups [1]. For A_n $(n \ge 3)$, B_n, D_n and D_4, the proofs in [6, 32] use an explicit description of $\mathrm{Aut}(A)$ or $\mathrm{Aut}(A/Z(A))$. For example, [6] embeds $\mathrm{Aut}(A/Z(A))$ in a mapping class group of finite index. For $I_2(m)$ the argument of [6] is different: it uses that $A/Z(A)$ is virtually free, hence non-elementary hyperbolic, and applies Fel'shtyn's theorem for hyperbolic groups.

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1.2. **Main result.** For a Coxeter graph Γ let $A = A(\Gamma)$ be the Artin group and $W = W(\Gamma)$ the Coxeter group.

Main Theorem. *The Artin groups of types E_6, E_7, E_8, F_4, H_3 and H_4 have property R_∞.*

To our knowledge $\mathrm{Aut}(A)$ has not been determined for any of these six types (cf. the list in [30, §1]). For F_4, Jones–Mangioni–Sartori [25, §4] record, citing Charney–Crisp [9, Prop. 9], that $\mathrm{Aut}(A(F_4))$ is virtually a direct product of infinite groups. Our proof avoids $\mathrm{Aut}(A)$ altogether.

Corollary. *Every irreducible Artin group of spherical type and rank ≥ 2 has property R_∞.*

Proof (from the literature and the Main Theorem). The irreducible spherical types of rank ≥ 2 are A_n $(n \geq 2)$, B_n $(n \geq 2)$, D_n $(n \geq 4)$, $E_{6,7,8}$, F_4, H_3, H_4 and $I_2(m)$ $(m \geq 5)$.

- $A_n, B_n, D_4, I_2(m)$ are covered by [6], and A_n also by [16] and, independently, by Proposition 5.1 below.
- D_n for $n \geq 6$ is covered by [32, Thm. 1].
- D_5 is credited to [32] in [31, §1], using the description of $\mathrm{Aut}(A(D_5))$ in [30]. The arXiv version v2 of [32] that we consulted still states D_5 as a conjecture, and [30] itself does not discuss R_∞. An independent proof for D_5 is given in Proposition 5.2.
- The remaining types are covered by the Main Theorem. □

Rank one must be excluded: $A(A_1) \cong \mathbb{Z}$, and inversion on $\mathbb{Z}$ has Reidemeister number 2.

1.3. **What is new, and relation to general criteria.** The new content is the Main Theorem and the method, which needs no knowledge of $\mathrm{Aut}(A)$. Section 5 applies the same criterion to other types:

- all A_n, by a proof by hand;
- B_n for $n \leq 7$ and D_n for $n \leq 7$, including D_5, by finite computation;
- some dihedral types $I_2(m)$; for others we show that the criterion fails.

None of these is a new result. Section 6 gives a uniform proof that the inversion automorphism has infinite Reidemeister number. This is not new in general: it follows from [31, Thm. 1] for non-spherical types; see also [26].

Conjectures (R) and (NA). Fel'shtyn and Troitsky asked whether every finitely generated residually finite group either has R_∞ or is solvable-by-finite [19]. This is Conjecture (R) in [32, §6]; see [27, Problem 19.28]. A weaker variant is Conjecture (NA): every finitely generated, residually finite, non-amenable group has R_∞. The statement (NA) appears as a theorem in the preprint [18], which would imply our Main Theorem. However, Soroko and Vaskou report that Troitsky has informed them that the proof contains a mistake [32, §6], so (NA) is currently open. Spherical Artin groups of rank ≥ 2 are finitely generated and linear, hence residually finite [11, 15], and they contain non-abelian free subgroups. Our results therefore verify Conjectures (R) and (NA) for these groups.

Acylindrical and quasimorphism criteria. Several recent criteria deduce R_∞ from a suitable action of $\mathrm{Aut}(G)$, or of $\langle \mathrm{Inn}(G), \varphi \rangle$, on a hyperbolic space, or from Aut-invariant quasimorphisms [20, 21, 33]. For an irreducible spherical Artin group A, the central quotient $A/Z(A)$ is acylindrically hyperbolic [7]. Applying these criteria still requires control over $\mathrm{Aut}(A)$ or $\mathrm{Aut}(A/Z(A))$, for instance an extension of the Calvez–Wiest action to the automorphism group. This control is not available for types E, F, H; see [25, Remark 4.8 and footnote 3] for a suggestion in this direction.

The R_∞ result [25, Cor. H] does not apply to spherical types, for two reasons. It needs a separating vertex of the defining graph, in which two generators are joined whenever $m_{st} < \infty$; for spherical types this graph is complete. It also needs a trivial centre.

Method and precedents. Forcing eigenvalue 1 on a characteristic abelian section by means of the representation theory of a reflection group goes back to Dekimpe–Gonçalves–Ocampo [12]. They used $\Gamma_2(P_n)/\Gamma_3(P_n)$ and Stembridge's results on eigenvalues [35].

The group $A/[P,P]$ is the "crystallographic quotient" studied in [23] and [28]. Crystallographic quotients of virtual Artin groups and their twisted conjugacy are studied by Dhanwani–Kumar–Naik–Singh [14]. The map f of §2.3 embeds $A/[P,P]$ into $\mathbb{Z}^{\mathcal{T}} \rtimes W$ (Remark 2.8). The action of A on $\mathbb{Z} \times \mathcal{T}$ that underlies f already appears in Cohen–Paris [10, Lemma 3.7].

We did not find these ingredients combined to prove R_∞ for spherical Artin groups in the literature. Our literature search was limited, since several bibliographic APIs were rate-limited, and we would welcome references.

1.4. **Outline of the proof.** Let $P = \ker(A \to W)$ be the pure Artin group, let $\mathcal{T} \subset W$ be the set of reflections, and let $\rho\colon W \to \mathrm{Sym}(\mathcal{T})$ be the conjugation action. Let $\varphi \in \mathrm{Aut}(A)$.

(1) $\varphi(P) = P$, by Cohen–Paris [10].
(2) $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$ as A-modules, where A acts through W by ρ, with no signs (Proposition 2.6). Hence $M_\varphi := \varphi_* \in GL(\mathbb{Z}^{\mathcal{T}})$ satisfies $M_\varphi \rho(w) M_\varphi^{-1} = \rho(\varphi_W(w))$.
(3) Suppose a permutation matrix π realises the automorphism of $\rho(W) \cong W/Z(W)$ induced by φ. Then $N = \pi^{-1} M_\varphi$ commutes with $\rho(W)$. By Schur's lemma, N acts as a rational scalar on a rational, absolutely irreducible, multiplicity-one constituent $U \subset \mathbb{Q}^{\mathcal{T}}$. Since N preserves the lattice $U \cap \mathbb{Z}^{\mathcal{T}}$, the scalar is ± 1.
(4) After composing φ with a suitable inner automorphism, M_φ has eigenvalue 1 on $H_1(P;\mathbb{Q})$. The eigenvector lies in U, or in $U_1 \oplus U_2$ in a swap variant used for F_4. Hence $R(\varphi|_P) = \infty$, and the finite-index inequality gives $R(\varphi) = \infty$.

The automorphisms of $W/Z(W)$ and the constituents U are determined by finite computations, which we certify by exact arithmetic (§4, Appendix A). For E_8 the automorphism group of $W/Z(W)$ is determined by hand (Proposition 4.2).

2. Preliminaries

2.1. **Reidemeister numbers.**

Lemma 2.1. *Let G be a group and $\psi \in \mathrm{Aut}(G)$.*

(a) If $K \trianglelefteq G$ and $\psi(K) = K$, let $\bar\psi$ be the induced automorphism of G/K. Then $R(\psi) \geq R(\bar\psi)$.
(b) If $G \cong \mathbb{Z}^m$ and ψ is given by $B \in GL_m(\mathbb{Z})$, then $R(\psi) = |\operatorname{coker}(I - B)|$. In particular $R(\psi) = \infty$ if B has eigenvalue 1.
(c) For $g \in G$ let $\iota_g(x) = gxg^{-1}$. Then $R(\iota_g \circ \psi) = R(\psi)$.
(d) If $H \leq G$ has finite index and $\psi(H) = H$, then $R(\psi|_H) \leq [G:H] \cdot R(\psi)$. In particular, if $R(\psi|_H) = \infty$ then $R(\psi) = \infty$.

Proof. (a) The projection $p\colon G \to G/K$ satisfies $p \circ \psi = \bar\psi \circ p$. Hence p maps the class of x into the class of $p(x)$, and it is surjective.

(b) Here y is ψ-conjugate to x iff $y - x \in (I - B)\mathbb{Z}^m$. If $\det(I - B) = 0$, the cokernel has positive rank.

(c) $y = zx(g\psi(z)^{-1}g^{-1})$ iff $yg = z(xg)\psi(z)^{-1}$. So right multiplication by g is a bijection from $\iota_g\psi$-classes to ψ-classes.

(d) Write $G = \bigsqcup_{i=1}^{k} Hg_i$ with $k = [G:H]$. Let $x \in H$, and suppose $y = zx\psi(z)^{-1} \in H$ for some $z \in G$. Write $z = hg_i$ with $h \in H$. Then $g_i x \psi(g_i)^{-1} = h^{-1} y \psi(h) \in H$, so y is $\psi|_H$-conjugate to $x_i := g_i x \psi(g_i)^{-1}$. Thus the ψ-class of x meets H in at most k $\psi|_H$-classes. Every $\psi|_H$-class lies in a ψ-class, so $R(\psi|_H) \leq k R(\psi)$. □

2.2. **Artin groups, Coxeter groups, reflections.** Let Γ be a Coxeter graph with vertex set S and Coxeter matrix (m_{st}). The Artin group is

$$A = \langle S \mid \Pi(s,t;m_{st}) = \Pi(t,s;m_{st}) \ (s \neq t,\ m_{st} < \infty)\rangle,$$

where $\Pi(s,t;m) = sts\cdots$ has m letters. The Coxeter group is $W = A/\langle\langle s^2 : s \in S\rangle\rangle$. Write $\mu\colon A \to W$, $a \mapsto \bar{a}$, and $P = \ker\mu$, the normal closure of $\{s^2 : s \in S\}$. Let

$$\mathcal{T} = \{wsw^{-1} : w \in W,\ s \in S\}$$

be the set of reflections, and let $\rho\colon W \to \mathrm{Sym}(\mathcal{T})$, $\rho(w)(r) = wrw^{-1}$. We write $\rho(w)$ also for the corresponding permutation matrix on $\mathbb{Z}^{\mathcal{T}}$, so $\rho(w)e_r = e_{wrw^{-1}}$.

Lemma 2.2. *We have* $\ker\rho = Z(W)$*, so* $\rho(W) \cong W/Z(W)$*. Every automorphism* α *of* W *preserves* $Z(W)$ *and induces an automorphism* θ_α *of* $\rho(W)$ *with* $\theta_\alpha(\rho(w)) = \rho(\alpha(w))$*.*

Proof. $\rho(w) = 1$ iff w commutes with every reflection. Since $\mathcal{T}$ generates W, this holds iff $w \in Z(W)$. □

For finite irreducible W of rank ≥ 2 we have $Z(W) \subseteq \{\pm 1\}$, and $-1 \in W$ exactly for types A_1, B_n, D_{2k}, E_7, E_8, F_4, H_3, H_4 and $I_2(2k)$ [24]. In rank ≥ 3 the map $r \mapsto \rho(r)$ is injective on $\mathcal{T}$. Indeed, if $\rho(r) = \rho(r')$ with $r \neq r'$, then $r' = -r$, which is not a reflection in dimension ≥ 3.

Theorem 2.3 (Cohen–Paris [10, Thm. 1.4])**.** *If* A *is an irreducible Artin group of spherical type, then* $P = \ker(A \to W)$ *is a characteristic subgroup of* A*.*

The statement of [10, Thm. 1.4] reads: "Let A be an irreducible Artin group of spherical type. Then CA is a characteristic subgroup of A," where $CA = \ker(\mu)$. Franco–Paris [22] extend this to reducible spherical types.

Computer assistance in Theorem 2.3. For the types treated in this paper, the proof in [10] is computer-assisted:

- For H_4, F_4, E_6, E_7, E_8 it rests on GAP calculations: all epimorphisms $A \to W$ are ordinary [10, Prop. 5.4, Cor. 5.5].
- For H_3 it uses a GAP calculation [10, Lemma 5.1] followed by an argument by hand [10, Prop. 5.3].

We have not re-run these calculations. The Main Theorem therefore also depends on them.

By Theorem 2.3, every $\varphi \in \mathrm{Aut}(A)$ satisfies $\varphi(P) = P$. Hence φ induces an automorphism φ_W of $W = A/P$, and an automorphism $\theta_\varphi := \theta_{\varphi_W}$ of $\rho(W)$.

2.3. **The homomorphism f and the abelianisation of P.** Let W act on $\mathbb{Z}^{\mathcal{T}} = \bigoplus_{r\in\mathcal{T}} \mathbb{Z}e_r$ by $w \cdot e_r = e_{wrw^{-1}}$, and form $\mathbb{Z}^{\mathcal{T}} \rtimes W$ with $(v,w)(v',w') = (v + w\cdot v', ww')$.

Lemma 2.4. *Let* W *be any Coxeter group. There is a unique homomorphism* $f\colon A \to \mathbb{Z}^{\mathcal{T}} \rtimes W$ *with* $f(s) = (e_s, s)$ *for* $s \in S$*. Its second component is* μ*. For a word* $a = s_1^{\epsilon_1}\cdots s_\ell^{\epsilon_\ell}$*,*

$$f(a) = \Big(\textstyle\sum_{k=1}^{\ell} \epsilon_k\, e_{r_k},\ \bar{a}\Big), \qquad r_k = \bar{s}_1\cdots\bar{s}_{k-1}\, s_k\, (\bar{s}_1\cdots\bar{s}_{k-1})^{-1}. \tag{1}$$

Proof. $(e_s,s)^{-1} = (-s\cdot e_s, s) = (-e_s, s)$, which gives (1) by induction on ℓ. We check the braid relations. Fix $s \neq t$ with $m = m_{st} < \infty$, and let $u = \Pi(s,t;m)$, $u' = \Pi(t,s;m)$.

- The W-components of $f(u)$ and $f(u')$ agree, because the braid relation holds in W.
- In W, st has order exactly m [24]. By (1), the $\mathbb{Z}^{\mathcal{T}}$-component of $f(u)$ is $\sum_{k=1}^{m} e_{r_k}$. Using $s(ts)^j = (st)^j s$, a direct computation gives $r_k = (st)^{k-1}s$ for both parities of k.
- For u' the reflections are $r'_k = (ts)^{k-1}t = (ts)^k s = (st)^{m-k}s$.
- Hence both $\{r_k\}$ and $\{r'_k\}$ equal $\{(st)^j s : 0 \leq j < m\}$, a set of m distinct elements of $\mathcal{T}$.

So $f(u) = f(u')$. □

Remark 2.5. Lemma 2.4 is equivalent to [10, Lemma 3.7(1)]. There, the action $U_s(k,t) = (k + \delta_{t,s}, sts)$ on $\mathbb{Z} \times \mathcal{T}$ commutes with translations, which is the same as a homomorphism to $\mathbb{Z}^{\mathcal{T}} \rtimes \mathrm{Sym}(\mathcal{T})$. The computation is also checked by machine (Appendix A).

Proposition 2.6. *Let A be any Artin group of spherical type, irreducible or not. Then $f(P) = 2\mathbb{Z}^{\mathcal{T}}$, and $f|_P$ induces an isomorphism $H_1(P;\mathbb{Z}) \to 2\mathbb{Z}^{\mathcal{T}}$. Under this isomorphism, conjugation by $a \in A$ acts on $H_1(P;\mathbb{Z})$ by $\rho(\bar{a})$, a permutation of coordinates with no signs.*

Proof. Since $\mu(P) = 1$, f maps P into the abelian group $\mathbb{Z}^{\mathcal{T}} \times \{1\}$, so $f|_P$ factors through $H_1(P;\mathbb{Z})$.

- $f(s^2) = (e_s + s \cdot e_s, 1) = (2e_s, 1)$, and $f(as^2a^{-1}) = f(a)(2e_s,1)f(a)^{-1} = (2e_{\bar{a}s\bar{a}^{-1}}, 1)$.
- P is generated by the conjugates of the s^2, and every reflection is conjugate to a simple one. Hence $f(P) = 2\mathbb{Z}^{\mathcal{T}}$.
- The same computation, $f(apa^{-1}) = \bar{a} \cdot f(p)$, gives the equivariance.

For injectivity: by Brieskorn [3], $P \cong \pi_1(X)$, where $X = \mathbb{C}^n \setminus \bigcup_H H_{\mathbb{C}}$ is the complement of the complexified reflection arrangement, with one hyperplane for each reflection. The first homology of the complement of a central arrangement of N complex hyperplanes is free abelian of rank N, generated by meridians [4, 29]. Thus $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{|\mathcal{T}|}$. The surjection $H_1(P;\mathbb{Z}) \to 2\mathbb{Z}^{\mathcal{T}} \cong \mathbb{Z}^{|\mathcal{T}|}$ is then an isomorphism, since finitely generated free abelian groups are Hopfian. □

Remark 2.7 (Signs)**.** Under $P = \pi_1(X)$, the element as^2a^{-1} is a meridian around the hyperplane of $\bar{a}s\bar{a}^{-1}$. The deck group W acts by complex-linear maps, which preserve the complex orientation of normal discs. This agrees with Proposition 2.6. As a further check, Δ^2 maps to $2\sum_r e_r$; this is verified by machine for the six types.

Remark 2.8 (Crystallographic quotient)**.** Since $\ker f \subseteq P$ and $\ker(f|_P) = [P,P]$, f embeds $A/[P,P]$ into $\mathbb{Z}^{\mathcal{T}} \rtimes W$ with image of index $2^{|\mathcal{T}|}$. The image contains the lattice $2\mathbb{Z}^{\mathcal{T}}$, with quotient W. This is the crystallographic group of [23, 28].

Corollary 2.9. *Let $\varphi \in \mathrm{Aut}(A)$, A irreducible spherical, and let $M_\varphi \in GL(\mathbb{Z}^{\mathcal{T}})$ be the matrix of the induced automorphism of $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$. Then*

$$M_\varphi \, \rho(w) \, M_\varphi^{-1} = \rho(\varphi_W(w)) = \theta_\varphi(\rho(w)) \qquad (w \in W), \tag{2}$$

and $M_{\iota_a \circ \varphi} = \rho(\bar{a}) M_\varphi$ for $a \in A$. In particular, the permutation character $\chi_{\mathcal{T}}(g) = \#\{r \in \mathcal{T} : g(r) = r\}$ of $\rho(W)$ satisfies $\chi_{\mathcal{T}} \circ \theta_\varphi = \chi_{\mathcal{T}}$.

Proof. Apply φ to apa^{-1} and use Proposition 2.6. For the last claim, take traces in (2). □

2.4. Rational, absolutely irreducible constituents of multiplicity one.

Definition 2.10. A $\rho(W)$-invariant subspace $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is a *good constituent* if $U \otimes \mathbb{C}$ is irreducible and occurs with multiplicity one in $\mathbb{C}^{\mathcal{T}}$. Such a U is absolutely irreducible, and it is the full isotypic component of its isomorphism type.

Lemma 2.11. *Let U be a good constituent and let $N \in GL(\mathbb{Z}^{\mathcal{T}})$ commute with $\rho(W)$. Then $N(U) = U$ and $N|_U = \pm 1$.*

Proof. $N(U)$ is invariant and isomorphic to U. By multiplicity one, $N(U) = U$. Since U is absolutely irreducible, $N|_U \in \mathrm{End}_{\mathbb{Q}W}(U) = \mathbb{Q}$, say $N|_U = c \in \mathbb{Q}^\times$. The lattice $\Lambda = U \cap \mathbb{Z}^{\mathcal{T}}$ has full rank in U, and $N^{\pm 1}(\Lambda) \subseteq \Lambda$. Take $v \in \Lambda$ with coprime coordinates. Then $cv \in \mathbb{Z}^{\mathcal{T}}$ gives $c \in \mathbb{Z}$, and likewise $c^{-1} \in \mathbb{Z}$. Hence $c = \pm 1$. □

Lemma 2.12 (Certificates)**.** *Let $U \subseteq \mathbb{Q}^{\mathcal{T}}$ be a nonzero $\rho(W)$-invariant subspace.*

(i) $\mathrm{End}_{\mathbb{Q}W}(\mathbb{Q}^{\mathcal{T}})$ is spanned by the orbital matrices O_ω, one for each $\rho(W)$-orbit ω on $\mathcal{T} \times \mathcal{T}$. If every O_ω acts on U as a scalar, then U is a good constituent.

(ii) Let χ_U be the character of U. If $\langle \chi_U, \chi_U \rangle = 1$ and $\langle \chi_U, \chi_{\mathcal{T}} \rangle = 1$, then U is a good constituent.

Proof. (i) A matrix commutes with every permutation matrix $\rho(w)$ iff it is constant on $\rho(W)$-orbits of $\mathcal{T} \times \mathcal{T}$. Write $\mathbb{C}^{\mathcal{T}} = \bigoplus_i V_i \otimes \mathbb{C}^{m_i}$. Then $\mathrm{End}_{\mathbb{C}W}(\mathbb{C}^{\mathcal{T}}) = \prod_i M_{m_i}(\mathbb{C})$.

- The central idempotents act on $U_{\mathbb{C}}$ by 0 or 1 and sum to 1, so $U_{\mathbb{C}} = V_i \otimes L$ for a single i and some $0 \neq L \subseteq \mathbb{C}^{m_i}$.
- $M_{m_i}(\mathbb{C})$ acts on L by scalars, which forces $m_i = 1$.

(ii) The characters of rational representations are real. So $\langle \chi_U, \chi_U \rangle = 1$ means $U_{\mathbb{C}}$ is irreducible, and $\langle \chi_U, \chi_{\mathcal{T}} \rangle$ is its multiplicity in $\mathbb{C}^{\mathcal{T}}$. □

3. The criterion

Theorem 3.1. *Let A be an irreducible Artin group of spherical type with Coxeter group W. Suppose that for every $\varphi \in \mathrm{Aut}(A)$ there exists a permutation π of $\mathcal{T}$ such that*

$$\pi\, \rho(w)\, \pi^{-1} = \theta_\varphi(\rho(w)) \quad \text{for all } w \in W, \tag{3}$$

and at least one of the following holds.

(i) (stable case) *There is a good constituent U such that for each $\varepsilon \in \{1, -1\}$ there exist $h \in \rho(W)$ and $0 \neq v \in U$ with $h\pi v = \varepsilon v$.*

(ii) (swap case) *There are good constituents $U_1 \neq U_2$ with $\pi(U_1) = U_2$, such that for each $\varepsilon \in \{1, -1\}$ there exist $h \in \rho(W)$ and $0 \neq v \in U_1$ with $(h\pi)^2 v = \varepsilon v$.*

Then A has property R_∞.

The hypothesis only concerns the automorphisms θ_φ of $\rho(W)$ induced by $\mathrm{Aut}(A)$. All of these preserve $\chi_{\mathcal{T}}$ (Corollary 2.9). In practice we verify the hypothesis for all automorphisms of $\rho(W)$ that preserve $\chi_{\mathcal{T}}$, or for all automorphisms.

Proof. Let $\varphi \in \mathrm{Aut}(A)$, and let π be as in the hypothesis. By (2) and (3), $N := \pi^{-1} M_\varphi \in GL(\mathbb{Z}^{\mathcal{T}})$ commutes with $\rho(W)$.

Case (i). By Lemma 2.11, $N|_U = c \in \{\pm 1\}$. Choose h and $v \neq 0$ with $h\pi v = cv$, choose $a \in A$ with $\rho(\bar{a}) = h$, and put $\varphi' = \iota_a \circ \varphi$. By Corollary 2.9, $M_{\varphi'} v = h\pi N v = c\, h\pi v = v$. Note that $h\pi$ need not preserve U.

Case (ii). Let $N|_{U_i} = c_i \in \{\pm 1\}$. Choose h and $0 \neq v \in U_1$ with $(h\pi)^2 v = c_1 c_2 v$, choose a with $\rho(\bar{a}) = h$, and put $\varphi' = \iota_a \circ \varphi$ and $Q = h\pi$. Then $M_{\varphi'} = QN$ and $QU_1 = U_2$, so

$$M_{\varphi'}^2 v = QNQNv = c_1 c_2\, Q^2 v = v.$$

Let $u = v + M_{\varphi'} v$. Then $M_{\varphi'} u = u$. Moreover $u \neq 0$: $M_{\varphi'} v \in U_2$, and $U_1 \cap U_2 = 0$ because distinct good constituents are distinct isotypic components.

In both cases $M_{\varphi'}$ has eigenvalue 1, hence a nonzero fixed vector in $\mathbb{Z}^{\mathcal{T}}$. By Lemma 2.1(b), Proposition 2.6 and Lemma 2.1(a) (with $[P, P]$ characteristic in P), we get $R(\varphi'|_P) = \infty$. Then $R(\varphi') = \infty$ by Lemma 2.1(d), since $[A : P] = |W|$, and $R(\varphi) = \infty$ by Lemma 2.1(c). □

Remark 3.2 (Reduction to outer classes)**.** Suppose π satisfies (3) for θ, and let $g \in \rho(W)$. Then $g\pi$ satisfies (3) for $\iota_g \circ \theta$. Since $\{hg\pi : h \in \rho(W)\} = \{h\pi\}$, conditions (i) and (ii) for $g\pi$ and for π are equivalent. Hence it suffices to check one representative of each class in $\mathrm{Out}(\rho(W))$ that can arise.

For the trivial class take $\pi = 1$. Condition (i) then holds as soon as U is a good constituent on which some reflection acts nontrivially: $h = 1$ gives eigenvalue $+1$, and a reflection acting nontrivially gives eigenvalue -1.

Corollary 3.3. *Suppose every automorphism θ_φ ($\varphi \in \mathrm{Aut}(A)$) is inner, and $\mathbb{Q}^{\mathcal{T}}$ has a good constituent on which some simple reflection acts nontrivially. Then A has property R_∞.*

Remark 3.4 (Permutations from reflection-preserving automorphisms)**.** Let θ be an automorphism of $\rho(W)$ with $\theta(\rho(\mathcal{T})) = \rho(\mathcal{T})$, and assume $r \mapsto \rho(r)$ is injective on $\mathcal{T}$. Then $\pi(r) := \rho^{-1}\theta(\rho(r))$ satisfies (3). Indeed, $\pi\rho(w)\pi^{-1}(r) = \rho^{-1}\big(\theta(\rho(w))\rho(r)\theta(\rho(w))^{-1}\big) = \theta(\rho(w))(r)$.

Lemma 3.5 (Reflection-preserving automorphisms, simply-laced case)**.** *Let Φ be an irreducible root system of type A_n, D_n or E_n (rank ≥ 3), with simple roots $\alpha_1, \dots, \alpha_n$ and Weyl group W. Let θ be an automorphism of $\rho(W)$ with $\theta(\rho(\mathcal{T})) = \rho(\mathcal{T})$. Then there are $w \in W$ and a diagram automorphism γ such that $\theta(\rho(s_i)) = \rho(w)\rho(s_{\gamma(i)})\rho(w)^{-1}$ for all i. In particular θ is inner for E_8 and E_7, and for E_6 and A_n, where the nontrivial diagram automorphism is induced by $-w_0$.*

Proof. By injectivity, $\theta(\rho(s_i)) = \rho(s_{\beta_i})$ for a root β_i, unique up to sign.

Orders of products. For roots $\beta \neq \pm\gamma$, the product $s_\beta s_\gamma$ is a rotation in a plane and the identity on its orthogonal complement. In rank ≥ 3, no nontrivial power of it lies in $Z(W) \subseteq \{\pm 1\}$. So its order in $\rho(W)$ equals its order in W: it is 2 iff $\beta \perp \gamma$, and 3 iff $(\beta, \gamma) = \pm 1$ (all roots have norm 2). Hence $(\beta_i, \beta_j) = 0$ when $m_{ij} = 2$, and $(\beta_i, \beta_j) = \pm 1$ when $m_{ij} = 3$.

Choice of signs. The Dynkin diagram is a tree, so we may change the signs of the β_i one vertex at a time, starting from a leaf, so that $(\beta_i, \beta_j) = -1$ on every edge. Then the Gram matrices of (β_i) and (α_i) coincide, and the linear map g with $g\alpha_i = \beta_i$ is an isometry.

g preserves Φ. The group gWg^{-1} is generated by $gs_{\alpha_i}g^{-1} = s_{\beta_i} \in W$. Hence $g\Phi = gW\{\alpha_i\} = (gWg^{-1})\{\beta_i\} \subseteq \Phi$, and $g\Phi = \Phi$ since both are finite of the same size. So $\{\beta_i\} = g\{\alpha_i\}$ is a simple system of Φ.

Conclusion. Since W acts transitively on simple systems [24], there is $w \in W$ with $w^{-1}g\{\alpha_i\} = \{\alpha_i\}$. The isometry $w^{-1}g$ permutes the simple roots and preserves their Gram matrix, so it induces a diagram automorphism γ. Then $\theta(\rho(s_i)) = \rho(gs_ig^{-1}) = \rho(w)\rho(s_{\gamma(i)})\rho(w)^{-1}$. □

The same argument works for B_n ($n \geq 3$) provided θ preserves each of the two reflection classes. For $n \geq 3$ this is automatic, since the classes have different sizes n and $n(n-1)$. Edges with $m_{ij} = 4$ join a long and a short root, and the order 4 forces $(\beta_i, \beta_j) = \pm 1$. The diagram of B_n has no automorphisms, so θ is inner.

4. Verification for $E_6, E_7, E_8, F_4, H_3, H_4$

4.1. Automorphisms of $W/Z(W)$.

Proposition 4.1. *Let $G = \rho(W) \cong W/Z(W)$.*

(a) For E_6 and E_7, $\mathrm{Out}(G) = 1$.
(b) For H_3, H_4 and F_4, $|\mathrm{Out}(G)| = 2$. The nontrivial class is represented by conjugation with the explicit permutation π_0 of $\mathcal{T}$ described below.
(c) For E_8, $\mathrm{Out}(G) = 1$ (Proposition 4.2; no computer needed).

Proof. (a), (b) are by computer (Appendix A). The following programs were used.

(1) `fastaut.py/pgrp.py` (original code) count all automorphisms by brute force over images of the Coxeter generators. The images are involutions in classes of the right size, subject to the Coxeter relations; the program checks that the map is a homomorphism and bijective. Results: $|\operatorname{Aut} G| = |G|$ for E_6, and $|\operatorname{Aut} G| = 2|G|$ for H_3, F_4 and H_4. This program was not run for E_7.
(2) `indep/out_check.py` (fresh code, exact roots) enumerates G, its involutions and their conjugacy classes. It checks that no involution class other than a reflection class has the size of a reflection class, so every automorphism maps $\rho(\mathcal{T})$ onto itself.

A generating tuple of reflections (t_i) with $\mathrm{ord}(t_it_j) = \mathrm{ord}(\rho(s_i)\rho(s_j))$ defines a surjection $W \to G$. Its kernel has order $|Z(W)| \leq 2$, so it is central and equals $Z(W)$. Hence such tuples correspond to automorphisms. The program counts their $\mathrm{Inn}(G)$-orbits. Results

agree with (1). For E_7 ($|G| = 1451520$) the involution classes have sizes $63, 315, 945, 3780$, and $\mathrm{Out}(G) = 1$.

(3) `referee_own_check.py`, written independently by the referee, recomputes $|\operatorname{Aut} G|$ for H_3 and F_4 without assuming that reflections are preserved.

For E_7 the computation of $\mathrm{Out}(G)$ thus rests on a single enumeration (`out_check.py`). It is corroborated in two ways:

- a stabiliser-chain recomputation (`out_check_stab.py`, same author) that reflection-preserving automorphisms are inner;
- Lemma 3.5, which proves the latter by hand.

The reflection-preservation step for E_7 was also recomputed by `uniform_check.py` (same author, separate code).

The permutations π_0.

- H_3, H_4. Let σ be the Galois automorphism $\sqrt{5} \mapsto -\sqrt{5}$, applied coordinatewise, and let τ swap the last two coordinates. Then $\Sigma = \tau \circ \sigma$ maps the root system Φ (coordinates in $\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$, see §A.1) onto itself. It is σ-semilinear and satisfies $(\Sigma x, \Sigma y) = \sigma((x, y))$, hence $\Sigma s_\alpha \Sigma^{-1} = s_{\Sigma\alpha}$. The induced permutation π_0 of $\mathcal{T}$ normalises $\rho(W)$, and $\pi_0^2 = 1$. The machine checks that no element of $\rho(W)$ induces the same automorphism, so it is outer.
- F_4. The map $L(x) = (x_1 + x_2, x_1 - x_2, x_3 + x_4, x_3 - x_4)$ is $\sqrt{2}$ times an orthogonal map. It sends short roots to long roots and long roots to twice short roots, so $Ls_\alpha L^{-1} = s_{L\alpha}$. The induced π_0 exchanges the two reflection classes, hence is outer, and $\pi_0^2 = 1$. □

Proposition 4.2. *Let $W = W(E_8)$ and $G = W/\{\pm 1\} \cong \rho(W)$. Every automorphism of G is inner.*

Proof. Step 1: the commutator subgroup. All simple reflections are conjugate, because the diagram is connected and simply laced. So W^{ab} is cyclic of order ≤ 2, and $\det\colon W \to \{\pm 1\}$ shows $W^{ab} = \mathbb{Z}/2$ with $[W, W] = W^+ := \ker \det$. Since $\det(-1) = 1$ in dimension 8, $-1 \in W^+$, and $G^+ := [G, G] = W^+/\{\pm 1\}$ has index 2 in G. Being the commutator subgroup, G^+ is characteristic. So every automorphism of G permutes the conjugacy classes of involutions in $G \setminus G^+$, preserving their sizes.

Step 2: involutions outside G^+. Let $\bar{g} \in G \setminus G^+$ be an involution with lift $w \in W$. Then $\det w = -1$ and $w^2 = \pm 1$. If $w^2 = -1$, the eigenvalues of w come in pairs $\pm i$, so $\det w = 1$, which is impossible. Hence $w^2 = 1$.

Let $V_-(w)$ be the (-1)-eigenspace of w, of dimension k, so $\det w = (-1)^k$ and k is odd. By Steinberg's theorem [34], the pointwise stabiliser of the fixed space V^w is generated by the reflections it contains, namely the s_β with $\beta \in V_-(w)$.

- By induction on k, w is a product of reflections in k mutually orthogonal roots spanning $V_-(w)$. Indeed, pick such a β; then ws_β is an involution with $V_-(ws_\beta) = V_-(w) \cap \beta^\perp$. (Compare [8, Lemma 5].)
- Replacing w by $-w$ replaces k by $8 - k$. So $\bar{g}$ is the image either of a reflection ($k = 1$), or of a product $s_{\beta_1} s_{\beta_2} s_{\beta_3}$ with $\beta_1, \beta_2, \beta_3$ mutually orthogonal ($k = 3$).

The second kind gives 37800 pairwise distinct elements of G. Indeed, w determines $V_-(w) = \langle \beta_1, \beta_2, \beta_3 \rangle$, and the only roots in this space are $\pm\beta_i$: a root $\sum c_i \beta_i$ has $c_i \in \{0, \pm\frac{1}{2}, \pm 1\}$ and $\sum c_i^2 = 1$. Moreover $-w$ has $k = 5$, so it is not of this form.

These elements form a single conjugacy class. The class of w corresponds to the W-orbit of the unordered orthogonal triple of root lines, and W acts transitively on ordered orthogonal triples of roots. To see this:

- W is transitive on the 240 roots [24].
- The roots orthogonal to a root form a root system of type E_7 (126 roots), whose Weyl group fixes the first root and is transitive on them.

- The roots of E_7 orthogonal to a root form a D_6 (60 roots), and $W(D_6)$ is transitive on them.
- The counts are readily checked from the coordinates of §A.1. This gives $240\cdot126\cdot60/(2^3\cdot3!) = 37800$ triples of lines in one orbit.

Thus the involutions in $G \setminus G^+$ form exactly two classes, of sizes 120 (the image $\rho(\mathcal{T})$ of the reflections, a single class) and 37800. Hence every automorphism θ of G satisfies $\theta(\rho(\mathcal{T})) = \rho(\mathcal{T})$.

Step 3. By Lemma 3.5 (type E_8 has no diagram automorphisms), θ is inner. □

Remark 4.3.
- The referee's program `referee_e8_involutions.py` confirms the class structure used in Step 2. It finds the $w^2 = 1$ involution classes of sizes $120, 1575, 3780, 37800, 56700$ (with $k = 1, 4, 2, 3, 4$) and one $w^2 = -1$ class of size 7560. The program `out_check_stab.py` independently confirms Step 3 by Schreier–Sims.
- The result agrees with the classical identification $W(E_8)/\{\pm1\} \cong O_8^+(2){:}2$ [2], but the proof above does not use it.
- For E_6 and E_7 the computer results agree with $W(E_6) \cong U_4(2){:}2$ and $W(E_7) \cong 2 \times S_6(2)$ [2]. In particular the E_7 involution class sizes $63, 315, 945, 3780$ are those of $S_6(2)$.

By Proposition 4.1 and Remarks 3.2 and 3.4, a permutation π as in (3) exists for every automorphism of $\rho(W)$, and in particular for every θ_φ: take $\pi = g$ or $\pi = g\pi_0$ with $g \in \rho(W)$.

4.2. **Good constituents and witnesses.**

Proposition 4.4. *Table 1 lists good constituents of $\mathbb{Q}^{\mathcal{T}}$.*

- *For E_6, E_7, E_8, H_3, H_4, every simple reflection acts nontrivially on each listed constituent.*
- *For F_4, only the long simple reflections s_1, s_2 act nontrivially on U_{long}, and only the short ones s_3, s_4 act nontrivially on U_{short}.*
- *For H_3 and H_4, π_0 preserves each listed U, and $\pi_0|_U$ has both eigenvalues $+1$ and -1. So (i) of Theorem 3.1 holds with $h = 1$ for both signs.*
- *For F_4, $\pi_0(U_{\text{short}}) = U_{\text{long}}$ and vice versa. Moreover $\pi_0^2 = 1$ has eigenvalue $+1$ on U_{short}, and $(s_1\pi_0)^2$ has eigenvalue -1 on U_{short}. So (ii) holds.*

type	$\lvert W\rvert$	$\lvert Z(W)\rvert$	$\lvert\mathcal{T}\rvert$	$\lvert\rho(W)\rvert$	$\lvert\operatorname{Out}\rho(W)\rvert$	$\mathbb{C}^{\mathcal{T}}$: good constituents; remainder
H_3	120	2	15	60	2	$1 + \mathbf{4}$; remainder $2\cdot5$
H_4	14400	2	60	7200	2	$1 + \mathbf{16} + \mathbf{25}$; remainder $\psi + \psi'$, $\dim 18$
F_4	1152	2	24	576	2	$2\cdot1 + \mathbf{2}_{\text{short}} + \mathbf{2}_{\text{long}} + 2\cdot9$
E_6	51840	1	36	51840	1	$1 + \mathbf{15} + \mathbf{20}$
E_7	2903040	2	63	1451520	1	$1 + \mathbf{27} + \mathbf{35}$
E_8	696729600	2	120	348364800	1	$1 + \mathbf{35} + \mathbf{84}$

TABLE 1. Data for the verification. Good constituents (bold) were certified exactly by `certify.py` (commutant criterion, Lemma 2.12(i)) and independently by `indep/indep_check.py` (character criterion, Lemma 2.12(ii)); for E_8 the second check uses the commutant criterion with fresh code. The remainders were determined as follows.

Remainders in Table 1.

- H_3: the remaining 10-dimensional part has $\langle\chi,\chi\rangle = 4$. Since the irreducible degrees of $\rho(W) \cong A_5$ are $1, 3, 3, 4, 5$, and four distinct ones sum to at least 11, it is $2\cdot5$.
- H_4: the remaining 18-dimensional part has $\langle\chi,\chi\rangle = 2$, so it is a sum of two non-isomorphic irreducibles ψ, ψ'. Numerically both have dimension 9 and are Galois conjugate.

- F_4: each of the two reflection classes (12 points each) carries a rank-3 action. There are 10 orbitals on $\mathcal{T} \times \mathcal{T}$: $3 + 3$ inside the two classes and $2 + 2$ between them. So each 12-dimensional permutation module is $1 + 2 + 9$ with distinct constituents. Its 2-dimensional constituents are U_{short} and U_{long}, which are non-isomorphic. The two modules share exactly two constituents, since $\dim \mathrm{Hom} = 2$. These are therefore the trivial and the 9-dimensional ones (`referee_f4_blocks.py`, `analyze.py`).
- E_6, E_7, E_8: the orbital counts are 3 and the decomposition is multiplicity free.

Proof of Proposition 4.4. By exact computer calculation (Appendix A). Each U is the kernel over $\mathbb{Q}$ of an integer matrix in the commutant, so it is defined over $\mathbb{Q}$. It is certified by Lemma 2.12(i) (`certify.py`) and by Lemma 2.12(ii) (`indep_check.py`, except for E_8).

The eigenvalue statements are certified by exact rational kernels, both by `certify_outer.py` (for an abstractly computed outer automorphism) and by `indep_check.py` (for the explicit π_0). The referee's `referee_own_check.py` reproduces the H_3 and F_4 cases. □

Proof of the Main Theorem.

- E_6, E_7, E_8: $\mathrm{Out}(\rho(W)) = 1$ by Propositions 4.1 and 4.2. Apply Corollary 3.3 with any constituent from Table 1.
- H_3, H_4, F_4: $\mathrm{Out}(\rho(W))$ has order 2. The trivial class is handled by Remark 3.2, and the nontrivial class by π_0 and Proposition 4.4. Apply Theorem 3.1. □

5. The criterion for the other spherical types

Proposition 5.1 (Type A_n; independent proof of a known result)**.** *For $n \geq 2$, $A(A_n)$ has property R_∞. The proof uses no description of* $\mathrm{Aut}(A)$.

Proof. Here $W = \rho(W) = S_{n+1}$, $\mathcal{T}$ is the set of transpositions, and $\mathbb{Q}^{\mathcal{T}}$ is the permutation module on 2-subsets.

- *Good constituent.* By Young's rule, $\mathbb{Q}^{\mathcal{T}} \cong S^{(n+1)} \oplus S^{(n,1)} \oplus S^{(n-1,2)}$ for $n + 1 \geq 4$, and $\mathbb{Q}^{\mathcal{T}} \cong S^{(3)} \oplus S^{(2,1)}$ for $n = 2$. These Specht modules are absolutely irreducible over $\mathbb{Q}$ and pairwise non-isomorphic. A transposition acts nontrivially on $U = S^{(n,1)}$.
- *Automorphisms.* For $n + 1 \neq 6$ every automorphism of S_{n+1} is inner. For $n + 1 = 6$, an outer automorphism maps transpositions to triple transpositions. Then $\chi_{\mathcal{T}}((12)) = 7$ but $\chi_{\mathcal{T}}((12)(34)(56)) = 3$, so by Corollary 2.9 it is not of the form θ_φ.

So every θ_φ is inner. Apply Corollary 3.3. □

Proposition 5.2 (Finite verification for classical types)**.** *The hypothesis of Theorem 3.1 holds, and hence A has property R_∞ (all known), for B_n ($3 \leq n \leq 7$), D_n ($4 \leq n \leq 7$) and $I_2(m)$ ($m \in \{9, 15\}$). The same program also recomputes E_6, E_7, F_4, H_3, H_4 (E_8 is too large to enumerate here). In particular this gives an independent proof for D_5.*

Proof. By exact computation with `indep/uniform_check.py`. For each type the program does the following.

(1) It checks that every involution class of $\rho(W)$ with the size and $\chi_{\mathcal{T}}$-value of a reflection class is itself a reflection class. Hence every θ_φ preserves $\rho(\mathcal{T})$, by Corollary 2.9.
(2) It enumerates the reflection-preserving automorphisms modulo Inn.
(3) It builds π as in Remark 3.4.
(4) It certifies good constituents by characters.
(5) It verifies (i) or (ii) of Theorem 3.1 by exact kernels for every class.

For D_4 all six triality classes satisfy (i) or (ii). For even n, D_n has one outer class, the diagram automorphism, which satisfies (i). □

Remark 5.3 (Towards all B_n and D_n)**.** For B_n ($n \geq 3$) and D_n ($n \geq 4$) the following can be proved by hand.

- *A good constituent.* Write $u_{ij} = e_{[e_i+e_j]} - e_{[e_i-e_j]} \in \mathbb{Q}^{\mathcal{T}}$. Then $U^- = \mathrm{span}\{u_{ij}\}$, of dimension $\binom{n}{2}$, is good. As a module over the normal subgroup $\{\pm 1\}^n$ it is the sum of the characters ε_I with $|I| = 2$, each with multiplicity one, and these characters occur nowhere else in $\mathbb{Q}^{\mathcal{T}}$. By Clifford theory, U^- is induced from an extension of ε_{12} to its inertia group, hence irreducible. It is moved by a simple reflection.
- *Reflection-preserving automorphisms* are inner up to diagram automorphisms (Lemma 3.5 and the remark after it). The diagram involution of D_n is realised by the sign change of e_n, which has eigenvalues ± 1 on U^-.

The missing step for a uniform proof is (1) of Proposition 5.2 for all n: that every automorphism of $W/Z(W)$ induced by $\mathrm{Aut}(A)$ maps reflections to reflections. We have verified it only for $n \leq 7$.

Remark 5.4 (Where the criterion fails: dihedral types)**.** Let m be odd. Then $\rho(W) = D_m$ acts on the m reflections as on the vertices of an m-gon, and

$$\mathbb{Q}^{\mathcal{T}} \cong \bigoplus_{d|m} U_d, \qquad \dim U_d = \varphi_{\mathrm{Euler}}(d),$$

where U_d is $\mathbb{Q}$-irreducible and absolutely irreducible iff $\varphi_{\mathrm{Euler}}(d) \leq 2$, i.e. $d \in \{1, 3\}$. So a nontrivial good constituent exists iff $3 \mid m$.

For $m = 5, 7$, and generally for m prime ≥ 5, there is none. Schur's lemma then only gives that $N|_{U_m}$ is a unit of an order in the real cyclotomic field $\mathbb{Q}(\cos\frac{2\pi}{m})$, which has infinitely many units. So the hypotheses of Theorem 3.1 cannot be met, and Schur's lemma alone gives no eigenvalue information.

For even m, the map $r \mapsto \rho(r)$ is not injective, and we have not analysed the criterion. The computation finds no good constituent for $m = 6, 10, 18$. Of course R_∞ holds for all $I_2(m)$ by [6].

6. Complements

Remark 6.1 (Centre-fixing automorphisms are elementary)**.** Let A be irreducible spherical. Then $Z(A) = \langle z \rangle \cong \mathbb{Z}$ with $z = \Delta$ or Δ^2 [5, 13], so $\varphi(z) = z^{\pm 1}$. The image of Δ in $H_1(A; \mathbb{Z})$ is nonzero: its coordinate sum is $\ell(w_0) = |\mathcal{T}|$. So if $\varphi(z) = z$, the induced automorphism of $H_1(A; \mathbb{Z})$ has eigenvalue 1, and $R(\varphi) = \infty$ by Lemma 2.1(a,b). Only centre-inverting automorphisms need Theorem 3.1.

Proposition 6.2 (Inversion; not new in general)**.** *Let A be an Artin group that is not free abelian, and let $\iota(s) = s^{-1}$ for $s \in S$. Then $R(\iota) = \infty$.*

Proof. ι maps each defining relation to the inverse of the relation with both words reversed, which is again a defining relation, so ι is an automorphism.

Let f be as in Lemma 2.4 and $\kappa(v, w) = (-v, w)$, an automorphism of $\mathbb{Z}^{\mathcal{T}} \rtimes W$. Then $f \circ \iota = \kappa \circ f$, so f maps ι-classes into κ-classes. For $g = (u, x)$ and $y = (v, w)$,

$$gy\kappa(g)^{-1} = (u + xv + xwx^{-1}u,\ xwx^{-1}).$$

Fix $s, t \in S$ with $m_{st} \geq 3$. On elements with W-part s, twisted conjugation acts by $v \mapsto (1+s)u + xv$ with $x \in C_W(s)$. Put $D_v(r) = v_r - v_{srs}$. Then:

- $D_{(1+s)u} = 0$;
- $x \in C_W(s)$ permutes the values of D_v.

So $\max_r |D_v(r)|$ is an invariant of the class. For $a_k = t^{2k}s$, $f(a_k) = (2ke_t + e_s, s)$ and the invariant equals $2k$, since $sts \neq t$. □

Appendix A. Computational methods and certificates

All code is in `code/`; commands and expected outputs are in `code/README.md`. Arithmetic is exact (Python `fractions`, SymPy `DomainMatrix`) wherever a certificate is claimed. Floating point is used only to locate candidate eigenvalues and witnesses, which are then rechecked exactly.

A.1. **Root systems.** The original code (`roots.py`) works in floating point with rounding. The independent code (`indep/rootsys.py`) is exact, with the following simple roots.

- $E_{6,7,8}$: Bourbaki's $\alpha_1 = \frac{1}{2}(1,-1,\dots,-1,1)$, $\alpha_2 = e_1+e_2$, $\alpha_k = e_{k-2}-e_{k-3}$.
- F_4: e_2-e_3, e_3-e_4, e_4, $\frac{1}{2}(e_1-e_2-e_3-e_4)$.
- A_n, B_n, D_n: the standard simple roots.
- H_3: the roots are $\pm 2e_i$ and the even permutations of $(\pm\phi, \pm 1, \pm\phi^{-1})$.
- H_4: the roots are $\pm 2e_i$, $(\pm 1,\pm 1,\pm 1,\pm 1)$ and the even permutations of $(\pm\phi,\pm 1,\pm\phi^{-1},0)$.

For H_3 and H_4 the simple roots are found by search.

A.2. **Programs.**

(1) `certify.py` (original). Commutant certificate, Lemma 2.12(i). It uses exact kernels of $X-\lambda$ for a symmetric integer element X of the commutant, and checks that every orbital matrix acts by a scalar.
(2) `certify_outer.py` (original). Brute-force outer automorphism, π as in Remark 3.4, and exact witness kernels.
(3) `indep/indep_check.py` (fresh). Exact roots, character certificates, explicit π_0 and exact witnesses; commutant mode for E_8.
(4) `indep/out_check.py`, `indep/out_check_stab.py` (fresh). Proposition 4.1.
(5) `indep/uniform_check.py` (fresh). Proposition 5.2.
(6) `indep/referee_own_check.py`, `referee_e8_involutions.py`, `referee_f4_blocks.py`. Written by the referee, independently of the other programs.
(7) `check.py` (original). Numerical sanity checks.

A.3. **Runtimes.** Single-threaded on a shared machine. Each program takes under 15 s, except `fastaut.py E6` (≈ 2.4 min) and `uniform_check.py` (≈ 1 min in total). The E_7 enumerations use about 1.3 GB of memory.

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